

Biasing in Computer Systems

1 Biasing in Floating-Point Representation

Overview

Biasing in floating-point representation is used to represent both positive and negative exponents using only unsigned integers. This is essential for floating-point formats like IEEE 754, which separate numbers into three parts: **sign bit**, **exponent**, and **mantissa (fraction)**.

How Biasing Works

- **Exponent Field:** In floating-point numbers, the exponent is stored as an unsigned integer, but the actual exponent can be both positive and negative. To allow this, a **bias** is added to the exponent before storing it.
- **Bias** is a fixed value used to offset the exponent. The bias value is subtracted from the stored exponent to determine the actual exponent.

IEEE 754 Example

- **Single Precision (32-bit):**
 - 1 bit for the sign, 8 bits for the exponent, and 23 bits for the fraction.
 - **Bias = 127:** $\text{Exponent} = \text{Stored Exponent} - 127$
- **Double Precision (64-bit):**
 - 1 bit for the sign, 11 bits for the exponent, and 52 bits for the fraction.
 - **Bias = 1023:** $\text{Exponent} = \text{Stored Exponent} - 1023$

Example

- $\text{Stored Exponent} = 130$
- $\text{Bias} = 127$
- $\text{Actual Exponent} = 130 - 127 = 3$
- The value represented by this exponent would be $2^3 = 8$

Benefits of Biasing

- **Simplified Hardware Design:** The exponent can be treated as an unsigned integer, simplifying hardware logic for comparison and arithmetic operations.
- **Range of Exponents:** Biasing allows representation of both very large and very small numbers, including negative exponents, without needing a sign bit for the exponent.

2 Biasing in Signed Number Representation (Two's Complement)

Overview

In two's complement representation, the **most significant bit (MSB)** acts as the sign bit, but the system also uses biasing to efficiently represent both positive and negative numbers.

How Biasing Works

- **Two's Complement Representation:**
 - Negative numbers are represented by flipping all bits of the number and adding 1. This makes subtraction and addition of signed numbers easier.

Bias in Two's Complement

- The **MSB** represents the sign:
 - 0 for positive numbers
 - 1 for negative numbers
- For an n -bit two's complement number, the range of values is from -2^{n-1} to $2^{n-1} - 1$

Example (8-bit Two's Complement)

- The 8-bit two's complement system has a range from -128 to $+127$
- To represent a negative number:
 1. Convert the absolute value to binary: $5 \Rightarrow 00000101$
 2. Invert the bits (one's complement): 11111010
 3. Add 1: 11111011 (this is the two's complement representation of -5)

Benefits of Biasing in Two's Complement

- **Simplified Arithmetic:** Two's complement arithmetic allows addition, subtraction, and comparison of signed numbers without special logic for the sign bit.
- **Efficient Representation:** It efficiently uses all available bits, allowing for a broad range of values (positive and negative) without needing separate sign handling.

3 Summary of Key Concepts

Biasing in Floating-Point Representation

- **Purpose:** Used to handle both positive and negative exponents in floating-point numbers.
- **How it Works:** Bias is added to the exponent to ensure it is non-negative. The actual exponent is computed by subtracting the bias.
- **Example:** In single-precision, an exponent of 130 would represent $130 - 127 = 3$, representing $2^3 = 8$.

Biasing in Signed Number Representation (Two's Complement)

- **Purpose:** To efficiently represent both positive and negative numbers using a single binary format.
- **How it Works:** The MSB is used for the sign, and negative numbers are represented using two's complement (invert bits and add 1).
- **Example:** In an 8-bit system, the two's complement representation of -5 is 11111011.

Benefits of Biasing

- **Floating-Point Biasing:** Simplifies comparisons and arithmetic by using unsigned integers for exponents.
- **Two's Complement Biasing:** Simplifies signed number representation and arithmetic operations by leveraging the same binary system for both positive and negative numbers.